

AI Audit Appendix

GenAI Usage:

This project used Claude (Anthropic), ChatGPT 5.5, and VSCode Gemini Copilot as a development assistant for:

- Initial scaffolding of the game simulator – ChatGPT

Original Prompt:

Create a lightweight custom Python simulation environment for the stochastic card game Flip 7 to support future reinforcement learning experiments using tabular Q-learning. Use simple card representations (integers and strings) instead of complex object-oriented card classes. The environment should include deck creation, player state tracking, round scoring, duplicate-card bust logic, special card handling (Freeze, Draw 3, Second Chance, multipliers, and bonus cards), heuristic-driven opponent behaviors (Conservative, Greedy, and Balanced), and multiplayer round flow for 2–4 players. Implement the environment using a Gym-style structure with `reset()`, `step()`, and state representation concepts, but avoid requiring OpenAI Gym itself. The initial focus should be on round-level training episodes rather than full-game simulations to simplify reward assignment and accelerate reinforcement learning experimentation

- Debugging training loop logic
- Guidance on Q-learning architecture decisions
- Debugging Q-learning State Space
- Debugging Heuristic Baseline Evaluation logic
- Conceptual guidance on Q-Learning Design Decisions
- LaTeX and Overleaf formatting
- Report Writing Refinement

All technical design decisions, hyperparameter choices, reward function design, and analytical conclusions are my own work.

In regards to writing refinement, Claude was used throughout the report drafting process with an initial draft, follow up targeted questions from Claude, which was then refined and into IEEE-

formatted academic prose. All technical content, design decisions, analytical conclusions and observations originate from my own work and understanding.